

Foundation (Jalapeno)

- Q1.** Solve the system: $x^2 + y^2 = 4$, $x - y = 0$
(A) $(x, y) = (0, 0)$
(B) $(x, y) = (1, 1)$
(C) $(x, y) = (-1, -1)$
(D) $(x, y) = (2, 2)$
(E) No solution
- Q2.** A spaceship travels from Earth to Mars. Its distance d from Earth after t hours is given by $d = t^2 + 2t + 1$. If the spaceship's velocity is given by $v = t + 1$, find t when $d = 12$ and $v = 4$
(A) $t = 1$
(B) $t = 2$
(C) $t = 3$
(D) $t = 4$
(E) $t = 5$
- Q3.** The graph of the system $y = x^2 - 2x + 1$, $y = -x^2 + 4x - 3$ has how many solutions?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q4.** Solve the system: $y = x^2 + 1$, $y = 2x + 1$
(A) $(x, y) = (0, 1)$
(B) $(x, y) = (1, 2)$
(C) $(x, y) = (-1, 2)$
(D) $(x, y) = (1, 3)$
(E) $(x, y) = (-1, 0)$
- Q5.** The equation $x^2 + y^2 = 9$ represents a circle with center at
(A) $(0, 0)$
(B) $(1, 1)$
(C) $(-1, -1)$
(D) $(3, 3)$
(E) $(-3, -3)$
- Q6.** A lab experiment involves the motion of a particle. Its position s after t seconds is given by $s = t^2 - 4t + 5$. If the particle's velocity is given by $v = 2t - 4$, find t when $s = 5$ and $v = 0$
(A) $t = 1$
(B) $t = 2$
(C) $t = 3$
(D) $t = 4$
(E) $t = 5$
- Q7.** The system $x^2 + y^2 = 16$, $y = x^2 - 4$ has how many solutions?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q8.** Solve the system: $x^2 - 2x + 1 = y$, $y = x^2 - 4x + 3$
(A) $(x, y) = (0, -3)$
(B) $(x, y) = (1, 0)$
(C) $(x, y) = (-1, -2)$
(D) $(x, y) = (2, 3)$
(E) $(x, y) = (3, 2)$

Open Ended Questions (FRQ)

- Q9.** A planet orbits the Sun in an elliptical path. The equation of its orbit is given by $x^2 + 4y^2 = 16$. If the planet's velocity is given by $v = 2x + y$, find the planet's coordinates when $v = 4$
- Q10.** A scientist conducts an experiment involving a projectile. Its height h after t seconds is given by $h = -t^2 + 4t + 2$. If the projectile's velocity is given by $v = -2t + 4$, find the time when the projectile hits the ground

Chef's Tips

- Tip 1: Encourage students to use graphing calculators to visualize the systems of equations
- Tip 2: Remind students to check their solutions by plugging them back into the original equations
- Tip 3: Emphasize the importance of labeling axes and using proper units when graphing